INCH-POUND

MIL-H-53073B
20 May 1993
SUPERSEDING
USED-IN-LIEU-OF
MIL-H-0053073A(ME)
14 May 1992 and
MIL-H-53073
27 July 1987

MILITARY SPECIFICATION

HOSE ASSEMBLY, ELASTOMERIC:

FUEL DISCHARGE, COLLAPSIBLE, LIGHTWEIGHT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1.1 Scope. This specification covers collapsible, lightweight, elastomeric discharge hose assemblies.

1.2 Classification. The collapsible discharge hose assemblies are of the following sizes and types as specified (see 6.2):

Size (Nominal Inside)

4-inch
6-inch

Types

O - Without couplings
A - Cam-Locking type coupling
B - Single groove, snap lock couplings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: SATBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4720

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

VV-F-800	- Fuel Oil, Diesel.

MILITARY

MIL-H-775	- Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing); Associated Hardware Packaging Of.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-C-10387	- Couplings, Clamp, Pipe, with Bolts and Fuel Synthetic Rubber Gasket, for Pipe and Tube.
MIL-F-16884	- Fuel Navy Distillate.
MIL-G-53006	- Gasoline, Automotive, Unleaded.
MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

FEDERAL

FED-STD-595	- Colors Used in Government Procurement.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MS27021	- Coupling Half, Quick Disconnect, Cam-Locking Type, Male, Hose Shank, Type II.
MS27025	- Coupling Half, Quick Disconnect, Cam-Locking Type, Female, Hose Shank, Type VI.
MS27028	- Coupling Half, Quick Disconnect, Cam-Locking Type, Cap, Dust, Type IX.
MS27029	- Coupling Half, Quick Disconnect, Cam-Locking Type, Plug Dust, Type X.
MS27030	- Gasket, Coupling Half, Quick Disconnect, Cam and Locking Type.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 380 - Rubber Hose.
D 381 - Existent Gum in Fuels by Jet Evaporation.
D 412 - Rubber Properties in Tension.
D 413 - Rubber Property - Adhesion to Flexible Substrate.
D 471 - Rubber Property - Effect of Liquids.
D 518 - Rubber Deterioration - Surface Cracking.
D 910 - Aviation Gasolines.
D 975 - Diesel Fuel Oils.
D 1053 - Rubber Property - Stiffening at Low Temperatures: Flexible Polymers and Coated Fabrics.
D 1149 - Rubber Deterioration - Surface Ozone Cracking in a Chamber.
D 1655 - Aviation Turbine Fuels.
D 3389 - Coated Fabrics Abrasion Resistance (Rotary Platform, Double-Head Abrader).
D 3951 - Standard Practice for Commercial Packaging.
D 4814 - Automotive Spark-Ignition Engine Fuel.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.2.1 Material deterioration prevention and control. The hose assemblies shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in the desert, the tropics, or moderate environments.

3.2.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.2.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The parts incorporated in the hose assemblies may be newly fabricated from recovered materials to the maximum extent practicable, provided the hose assemblies produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the hose assemblies.

3.3 Design and construction. The hose assemblies shall consist of collapsible, reinforced elastomeric hose with attached cam-locking type coupling halves, or grooved end fittings.

3.3.1 Hose. The hose shall be constructed of an elastomeric inner tube, a synthetic fiber reinforcement, and an elastomeric cover.

3.3.1.1 Hose tube. The tube shall be fabricated from a synthetic compound. The compound shall be resistant to aromatic hydrocarbon fuels, and shall not have any deleterious effects upon or be degraded by the following military and commercial fuels:

MIL-G-3056, automotive gasoline, MOGAS
ASTM D 4814 automotive gasoline (allows use of oxygenates)
MIL-G-53006 gasohol (containing ethanol)
ASTM D 910 aviation gasoline
MIL-T-5624 turbine fuel, JP-4 and JP-5
MIL-T-83133 turbine fuel, JP-8
ASTM D 1655 turbine fuel, Jet A and Jet A-1
VV-F-800 diesel fuel, DFA, DF-1 and DF-2
MIL-F-16884, navy distillate fuel
ASTM D 975 diesel fuel, 1-D and 2-D

The tube shall be smooth, free from pits, and of uniform thickness. The thickness of the tube shall be not less than 1/32-inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

3.3.1.2 Hose cover. The cover shall be fabricated from a synthetic compound. The cover shall be uniform in thickness and shall be not less than 1/32-inch. Unless otherwise specified (see 6.2), the color of the cover shall approximate lusterless tan 686 in accordance with FED-STD-595, color chip no. 33446.

3.3.1.3 Hose reinforcement. The reinforcement material shall be one or more plies of woven braided or spiraled synthetic fiber cord or fabric which shall be evenly and firmly fabricated and thoroughly impregnated with synthetic compound which shall cause the plies to adhere firmly to each other and to the tube and cover. The reinforcement material shall be applied by weaving, braiding, or spiral winding twist to provide a balanced design. The reinforcement shall be evenly and firmly fabricated, and shall be free from defects, dirt, knots, lumps, and irregularities of twist so that the completed hose shall comply with all the applicable requirements herein.

3.3.1.4 Static bonding. The hose shall contain continuous embedded static wire(s) or other electric conductor(s). When tested as specified in 4.5.2.1, the conductor(s) shall provide a continuous electric bond between the fittings on the ends of the hose. The maximum resistance shall not exceed 1.5 ohms per linear foot of hose. The conductor(s) shall not break when the hose is coiled, sharply bent or subjected to internal pressure. Each field-applied end shall meet these requirements.

3.3.2 Coupling halves. The coupling halves shall be type A or B, as specified (see 1.2 and 6.2). Coupling halves shall be securely attached to the hose to form a permanent electric bond with the static bonding. Type O hose has no couplings.

3.3.2.1 Type A, cam-locking coupling. When specified (see 6.2), type a cam-locking type coupling halves with specified gaskets shall be furnished with both size of hoses as shown in table I, except that the working pressure rating shall be 150 psi (pounds per square inch). Dust caps, plugs, and gaskets shall be in accordance with MS27028, MS27029, and MS27030, respectively.

TABLE I. Coupling halves and clamps.

Hose Size Nominal (inches)	Couplings	
	Female	Male
4	MS27025-17	MS27021-17
6	MS27025-19	MS27021-21

3.3.2.2 Type B, single grooved couplings.

3.3.2.2.1 Grooved coupling. The grooved couplings shall conform to MIL-C-10387, class A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

3.3.2.2.2 Hose to grooved adapter. For type B couplings, a one piece aluminum adapter shall be provided with a grooved-end conforming to MIL-C-10387, figure 1. For both sizes, the dimensions shown in MIL-C-10387 for the adapters shall be used. The nipple shall conform to the dimensions shown on MS27021, for the "C" length only.

3.3.3 Hose clamps. Each cam-locking type coupling half and single grooved coupling shall be secured to the hose with bolted aluminum clamps. When installed and locked, the clamps shall have no protruding ends.

3.4 Physical and chemical properties.

3.4.1 Proof pressure. When tested as specified in 4.5.2.1, the hose with couplings attached shall not leak nor show any imperfections in either the hose or couplings.

3.4.1.1 Minimum burst. When tested as specified in 4.5.2.2, each hose shall have a minimum burst pressure of 600 psi. Hose shall not fail radially.

TABLE II. Burst and working pressure.

Size I.D. (inside diameter) (inches nominal)	Burst pressure psi (minimum)	Working pressure (psi)
4	600	150
6	600	150

3.4.2 Tensile strength. When tested as specified in 4.5.2.3.1, the tensile strength of the tube shall be not less than 1,250 pounds force per square inch (psi) and the tensile strength of the cover shall be not less than 1,000 psi. When tested as specified in 4.5.2.3.1, the tensile strength of the tube and cover after fuel immersion shall be not less than 40 percent or less than 600 psi (whichever is greater) of the tensile strength obtained before immersion.

3.4.2.1 Tensile strength after water immersion. When tested as specified in 4.3.3, the tensile strength of the tube and cover after 14-day water immersion shall be at least 40 percent of the tensile strength obtained before immersion and shall be at least 600 psi. The tensile strength after 42-day water immersion shall be at least 25 percent of the tensile strength before immersion and shall be at least 300 psi.

3.4.3 Ultimate elongation after water immersion. When tested as specified in 4.3.3, the ultimate elongation after 14-day water immersion, shall be not less than 40 percent of the original elongation obtained before immersion. The ultimate elongation, after 42 days, shall be not less than 25 percent of the original elongation obtained before immersion.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

3.4.4 Adhesion.

3.4.4.1 Original. When tested as specified in 4.5.2.3.2, the original adhesion between the tube and reinforcement, and between the cover and the reinforcement, and between the cover and end reinforcement, shall be not less than 4.45 decanewton (daN) for the cover and 4.45 daN for the lining.

3.4.4.2 After fill test. When tested as specified in 4.5.2.3.3, the adhesion between the tube and reinforcement, between the reinforcement, and between the cover and the reinforcement and between the cover and end reinforcement, shall be not less than 4 daN for the cover and 3.5 daN for the lining.

3.4.5 Volume increase. When tested as specified in 4.5.2.4, the volume increase of the tube specimen after immersion in test fluid shall not exceed 40 percent, and the volume increase of the cover specimen shall not exceed 70 percent.

3.4.6 Low temperature flexibility. When tested as specified in 4.5.2.5, the relative torsional modulus for the inner tube and the outer cover shall not be greater than 5 at -32 °C (-25.6 °F).

3.4.7 Existent gum content. When tested as specified in 4.5.2.6, the remainder shall be not more than 6 milligrams per 100 milliliters of test fluid.

3.4.8 Ozone resistance. When tested as specified in 4.5.2.7, the hose cover material shall show no evidence of cracking, checking, or irregularities at the end of the exposure time, when viewed under 7 power magnification.

3.5 Lengths, diameters, and weights. The length of the hose shall be as specified (see 6.2) and shall be the actual hose length exclusive of the couplings. A tolerance in length of ±2 percent shall be permitted. This hose shall be furnished in the diameters and weights shown in table III.

TABLE III. Diameters and weights.

Inside diameter (inches ±1/16 inch)	Maximum weight (Pounds per foot)
4-1/16	1.5
6-1/16	2.0

3.6 Identification marking. Each length of hose shall be marked with:

Manufacturer's name or trademark
Quarter and year of manufacture
Manufacturer's serial number
Working pressure
Hose, Liquid Petroleum Fuels, Dispensing, Collapsible

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

The hoses shall be branded with the manufacturer's standard working pressures for the specific type and size hose supplied, and shall be rated at 150 psi. The letters in the marking shall be molded, either recessed or in relief, or shall be hot foil printed. The letters shall be in a contrasting color to the hose color and shall be not less than 1/4-inch high. Hoses shall be marked at a maximum of 10-foot intervals or by continuous-impression branding, or hot foil printing provided that the pattern is repeated every 5 feet or less.

3.7 Abrasion resistance. When tested as specified in 4.5.2.8, the number of cycles needed to expose the reinforcing material shall be reported.

3.8 Workmanship. The hose assemblies shall have no cuts, nicks, bruises, blisters, sharp edges, or lumps, and shall be clean and smooth. Each hose assembly shall be free from any workmanship deficiencies that could impair the function or serviceability of the hose or end fitting.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

4.3 First article inspection.

4.3.1 Examination. The first article hose assembly shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article hose assembly shall be tested as specified in 4.5.2.1 through 4.5.2.5 inconclusive and 4.5.2.7, and 4.5.2.8. Failure of any test shall be cause for rejection of the hose assembly.

4.3.3 Tensile strength and ultimate elongation after water immersion. Tensile strength and ultimate elongation after water immersion shall be determined in accordance with ASTM D 471. The immersion test shall be performed in accordance with ASTM D 471, utilizing water as a test fluid. One specimen shall be subjected to water immersion test for 14 days at 23 °C. Nonconformance to 3.4.2.1 or 3.4.3 shall constitute failure of this test.

4.4 Quality conformation inspection.

4.4.1 Lot. Each lot shall consist of all hose assemblies of the same size manufactured under the same conditions and at the same time, and offered to the Government at one time.

4.4.2 Sampling. Sampling of hose and hose assemblies for examination and test shall be in accordance with MIL-STD-105, table I and table IIa minimum burst pressure test (see 4.5.2.2). A lot shall be accepted when 0 defects are found and rejected when 1 or more defects are found.

4.4.3 Examination. Samples selected in accordance with 4.4.2 shall be examined as specified in 4.5.1.

4.4.4 Tests.

4.4.4.1 Individual. Each hose assembly shall be tested as specified in 4.5.2.1. Failure of the test shall be cause for rejection.

4.4.4.2 Samples. Samples selected in accordance with 4.4.2 shall be tested as specified in 4.5.2.2 through 4.5.2.5, inclusive.

4.5 Inspection procedure.

4.5.1 Examination. The hose assembly shall be examined as specified herein for the following defects:

101. Materials not as specified (see 3.2).
102. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments (see 3.2.1).
103. Contractor does not have documentation available for identification of materials, material finishes, or treatment (see 3.2.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

104. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the hose assembly (see 3.2.3).
105. Design and construction not as specified (see 3.3).
106. Color of cover not as specified (see 3.3.1.2).
107. Hose clamp not as specified (see 3.3.3).
108. Couplings not as specified (see 3.3.2).
109. Diameters and weights not as specified (see table III).
110. Identification marking missing, illegible, or not as specified (see 3.6).
111. Workmanship not as specified (see 3.8).

4.5.2 Tests.

4.5.2.1 Proof test and electric bond. Each length of hose, with fittings such as couplings and adapters attached, shall be subjected to a 225 psi hydrostatic pressure test in accordance with ASTM D 380 for all sizes. The electric bond shall be determined while full test pressure is applied. Nonconformance to 3.3.1.4 and 3.4.1 shall constitute failure of this test. Electric resistance shall be measured between couplings and shall be not more than 1.5 ohms per linear foot of hose.

4.5.2.2 Minimum burst. A 3 to 6 foot long section shall be cut from either end of one length selected from each 100 lengths for burst and physical test. The hose shall be subjected to a hydrostatic test pressure in accordance with ASTM D 380 and as specified in table II. The pressure shall be supplied at a uniform rate sufficient to develop the test pressure in 120 seconds, ±2 seconds. Nonconformance to 3.4.1.1 shall constitute failure of this test. The remaining section of hose from which the test sample has been taken shall be accepted as a full length, provided the sample meets all the requirements of this specification.

4.5.2.3 Tensile strength, elongation, and adhesion.

4.5.2.3.1 Tensile specimens. Tensile specimens shall be tested in accordance with ASTM D 412 method A. Die "C" shall be used for cutting test specimens. The tube specimens shall be buffed to not less than 0.050-inch thickness. Nonconformance to 3.4.2 and 3.4.3 shall constitute failure of this test.

4.5.2.3.2 Original adhesion test. A dry specimen of hose material shall be tested in accordance with ASTM D 413 machine method. Nonconformance to 3.4.4.1 shall constitute failure of this test.

4.5.2.3.3 Adhesion after filling. A 12-inch length of the hose shall be stoppered and filled with ASTM D 471 reference fuel B. The section of hose with the test fluid shall be maintained at a temperature of 23 °C (160 ±2 °F) for not less than 46 hours. After cooling to 23 ±5 °C, ring specimens shall be cut from the 12-inch length of hose and tested in accordance with ASTM D 413 machine method within 15 minutes after the removal of the test fluid. Nonconformance to 3.4.4.2 shall constitute failure of this test.

Review activities: Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

4.5.2.4 Volume increase. Volume increase test shall be determined in accordance with ASTM D 471 utilizing reference fuel B at 23 °C for 46 hours. Nonconformance to 3.4.5 shall constitute failure of this test.

4.5.2.5 Low temperature flexibility. The test for low temperature flexibility shall be made in accordance with ASTM D 1053. The low temperature test shall be determined after the samples have been conditioned 7 days at -32 °C (-25.6 °F). Nonconformance to 3.4.6 shall constitute failure of this test.

4.5.2.6 Existent gum. A test sample of hose, not less than 14 inches long, shall be selected and the bottom plugged with a clean corrosion-resisting steel cylinder 2 inches long secured in place with a clamp. The sample shall be filled to within 2 inches of the top with reference fuel B. The top end of the hose shall be plugged in a manner similar to the bottom. The sample shall then be stored in a vertical position for 7 days at an ambient temperature of 38 ±1 °C (100 ±2 °F). At the end of each 24 hours, the fuel in the sample shall be agitated for 5 minutes by moving the hose back and forth from the vertical to a horizontal position at a rate of 2 cycles per minute. At the end of the 7-day storage period, the fuel shall again be agitated in the sample for 5 minutes and immediately removed from the hose. The removed fuel shall be tested for existent gum in accordance with the air-jet solvent wash method of ASTM D 381. A blank shall be run on the test fluid at the same time and by the same method. The existent gum of the blank fluid shall be subtracted from the existent gum obtained from the test fluid removed from the hose. Nonconformance to 3.4.7 shall constitute failure of this test.

4.5.2.7 Ozone resistance. Specimens of the cover, prepared as described in ASTM D 518, specimen A, shall be mounted in a 20 percent elongated position and tested in accordance with ASTM D 1149. After conditioning for 24 hours in an ozone-free atmosphere, the mounted specimens shall be exposed for 336 hours at 40 ±1 °C (104 ±1 °F), to an atmosphere containing 50 parts ±10 parts, per hundred million (pphm) of ozone. Upon completion of the test, the specimens shall be examined. Nonconformance to 3.4.8 shall constitute failure of this test.

4.6 Inspection of packaging. The packaging shall be examined to determine compliance with the quality assurance provisions of MIL-H-775 and with the requirements of section 5 herein.

5. PACKAGING

5.1 Preservation. Preservation shall be level C (non-weather resistant). The end fittings shall be wrapped to prevent their rubbing on the hose during shipping.

5.2 Packing. The hose assemblies shall be coiled and the coils secured. Packaging and marking shall be commercial in accordance with ASTM D 3951.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The hose assemblies covered by this specification are intended for transferring gasoline, jet fuel, and diesel fuels.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number, and date of the specification.
b. Size required (see 1.2).
c. Type of coupling (see 1.2 and 3.3.2).
d. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
e. When a first article is not required (see 3.2 and 4.3).
f. Color of cover, if other than as specified (see 3.3.1.2).
g. When type A, cam-locking couplings shall be furnished (see 3.3.2.1).
h. Length of hose (see 3.5).
i. Level of preservation and packing required (see 5.1 and 5.2).

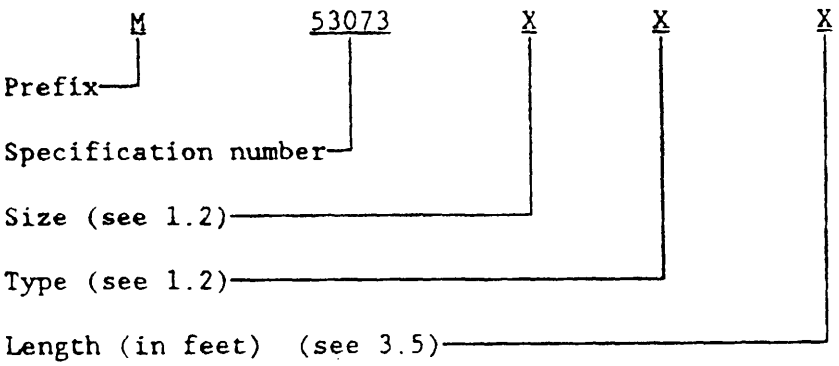
6.3 First article. When first article inspection is required, the items should be a preproduction model. The first article should consist of one hose assembly. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first article. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Subject term (key word) listing.

Collapsible
Discharge
Fuel
Hose

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

6.5 Part or identifying number (PIN). The PIN number to be used for hose assembly acquired by this document are as follows:



Example - M530734A500 (A hose assembly conforming to MIL-H-53073 having a 4-inch inside diameter with cam-locking type couplings and a length of 500 feet.)

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD Far Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title	Suggested Tailoring
4.3	DI-T-4902	First Article Inspection Report	See Below

The above DIDs were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

In addition to the results, from first article examination and tests listed in 4.3, the First Article Test Report should also include results from the following abrasion test.

6.3.1 Abrasion resistance. The contracting officer should require the contractor to perform an abrasion test as follows:

Cut three test samples from the hose with the abrasion testing performed on the elastomeric cover side. Perform the abrasion test in accordance with one of the following procedures:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

a. ASTM D3389, method A. using abrasion wheel H-22 having neither additional weight nor counterweight applied to the specimen holder.

b. ISO 4649, method A.

The method of testing used (i.e., ASTM D3389 or ISO 4649) and the number of cycles needed to expose the reinforcing material should be included in the first article test report. The results of this test are for information purposes only and should not be used as criterion for acceptance of hose assemblies.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activities:
Navy - SH
Air Force - 82
DLA - CS

User activity:
Navy - MC

Preparing activity:
Army - ME

Project 4720-0031

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-H-530738	2. DOCUMENT DATE (YYMMDD) 930520

3. DOCUMENT TITLE
Hose Assembly, Elastomeric: Fuel Discharge, Collapsible, Lightweight

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468	(2) AUTOVON 654-3468
ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: SATBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.